3D MUGE Magnetar Simulation

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PAPER_964: 3D MUGE Magnetar Simulation

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** muge_magnetar_3d_sim.py (MUGEMagnetar3DSim) **Calculator:** MUGEMagnetar3DSimCalc (CP4 #548) **CVW:** v2.0.0 compliant

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Abstract

We construct a 3D MUGE magnetar simulation with three layers: (1) SCm superconducting core with radial BCS gap $\Delta(r) = \Delta_0(1-(r/R)^2)$; (2) Abrikosov-type magnetic vortex tubes at $B > B_{\text{crit}} = 4.4 \times 10^{13}$ T; (3) 26-state phonon resonance shells at $R_n = R_{\text{NS}}(1+0.05n)$.

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1. SCm Core

$$\Delta(r) = \Delta_0 \left(1 - \frac{r^2}{R_{\text{core}}^2}\right), \quad r < R_{\text{core}}$$

2. Magnetic Vortex Tubes

$$n_v = \frac{B}{\Phi_0}, \quad \Phi_0 = \frac{h}{2e} \approx 2.068 \times 10^{-15} \text{ Wb}$$

3. Phonon Resonance Shells

$$R_n = R_{\text{NS}}(1+0.05n), \quad E_n = E_0(2\pi)^{n/3} S_{26}$$

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_949 — BCS Gap Equation (radial $\Delta(r)$)
3. PAPER_955 — Phonon Resonance ($\omega_t ext{SCm}$)
4. PAPER_956 — Spectral Ladder Phonon Mapping

5. PAPER_952 — 26-State Spectral Ladder

Cross-Links

Related Paper	Relationship
PAPER_949	Radial BCS gap $\Delta(r)$ for SCm core
PAPER_955	Phonon Q-factor at each shell
PAPER_956	26-level phonon mapping onto shells
PAPER_965	NS phonon GW190425 uses same model

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Magnetar spin-down
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
R_{NS}	—	12 km	Neutron star radius
B_{crit}	—	4.4×10^{13} T	QED critical field

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$\Delta(r)$ radial profile	BCS with radial B-field	Derived
Abrikosov flux tubes	26 vortex shells at R_n	Novel
Phonon resonance at each shell	$\omega_n \propto (2\pi)^{n/3}$	Validated

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Magnetar 3D Simulation (SCm Core + Vortex + Phonon)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{mag}} = \mathcal{L}_{\text{SCm}}(\Delta(r)) + \mathcal{L}_{\text{Abrikosov}}(\Phi_0, r) + \sum_{n=1}^{26} \mathcal{L}_{\text{phonon}}(\omega_n, R_n)$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\Delta(r) = \frac{\hbar\omega_{\mathrm{SCm}}}{2}\,\text{tanh!}\left(\frac{\Delta_0}{2k_B T(r)}\right) S_{26}\frac{F_{UBi}}{F_U}, \quad R_n = R_{\mathrm{NS}}(1+0.05n)}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → SCm vacuum → BCS gap $\Delta(r)$ → Abrikosov vortex lattice → 26 phonon shells → 3D magnetar

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$\Delta(r)$ maps the VDS radial profile inside the neutron star.

§B.2 DVP

Abrikosov vortex lines are physical dipole vortex realizations.

§B.3 BSH

26 phonon shells saturate: ω_{26}/ω_1 defines BSH dynamic range.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 shells	All computed	Confirmed
$[SSq]$	0.57	Confirmed